

Vignesh Udhayakumar

Experienced Backend Developer with nearly 6 years of hands-on experience in designing, developing, and maintaining scalable web applications. Highly proficient in Python and backend frameworks such as FastAPI, Django, and Flask. Strong expertise in AWS cloud architecture and services, with additional specialization in AI integrations using OpenAI APIs. Demonstrates a solid background in data science, machine learning model development, and advanced debugging. Adept at delivering high-quality code, optimizing performance, and collaborating in agile environments.

Work Experience

Gramener Technology Solutions Pvt Ltd

Senior Data Science Associate Engineer - Feb 2024 – Present

AI-Powered Web Monitoring & Data Extraction Platform

- Developed a robust web scraping system using Scrapy and Selenium to extract structured and unstructured data (text, images, PDFs) from U.S. government websites.
- Automated OCR and text recognition from visual content via AWS Textract and PyTesseract, significantly enhancing data ingestion accuracy.
- Integrated Azure OpenAI and Hugging Face LLMs to intelligently extract, summarize, and contextualize business-relevant insights from unstructured data sources.
- Designed a scheduled scraping engine with differential analysis, highlighting changes over time and surfacing deltas through a custom-built dashboard.
- Delivered state-level analytics, offering actionable insights derived from large-scale government content across all 50 U.S. states.
- Built scalable pipelines to process and validate large datasets (CSV/XLSX) using Pandas, aligning them to internal schemas using OpenAI models for semantic column mapping.
- Stored cleansed data in PostgreSQL, with backend workflows orchestrated through AWS Lambda and DynamoDB, supporting a fully serverless architecture.

Tech Stack: Python, Scrapy, Selenium, AWS, OpenAI, Hugging Face, PostgreSQL

Concertante Labs LLP

Senior Software Engineer - Sep 2019 - Jan 2024

- Led the end-to-end development of ConcertoAI, an intelligent chatbot designed to automate and streamline front desk operations, reducing manual workload and response time.
- Implemented core features such as appointment scheduling, rescheduling, and cancellation, providing 24/7 client access and improving service continuity outside business hours.
- Integrated LLM-based conversational AI (OpenAI) and custom workflows to handle complex inquiries, resulting in faster and more consistent client engagement.
- Trained the chatbot with a comprehensive knowledge base, enabling accurate responses to frequently asked questions and policy-specific queries.
- Architected backend using FastAPI and deployed services on AWS, ensuring scalability and high availability for real-time interactions.

Tech Stack: Python, MongoDB, FASTAPI, Jenkins, Docker, Redis

Skills

Technical Skills

- Languages: Python, SQL, Bash
- Web Frameworks: FastAPI, Django, Flask
- APIs: REST, GraphQL (basic), OpenAI, Rasa
- Web Scraping: Scrapy, Selenium, BeautifulSoup
- OCR & Parsing: AWS Textract, PyTesseract
- Databases: PostgreSQL, MySQL, SQLite, DynamoDB
- Data Science: Pandas, NumPy, scikit-learn, TensorFlow
- LLMs & NLP: OpenAI GPT, Hugging Face, LangChain
- Cloud: AWS (EC2, S3, Lambda, API Gateway), Azure (OpenAI)
- DevOps: Docker, Git, CI/CD, Terraform
- Scripting: Python scripts, Bash automation
- Testing & Debugging: PyTest, profiling, log tracing

Contact

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Educational Background

B.Sc - Computer Science

Thiruvalluvar University

Jun 2013 - May 2016

- Specialization in Computer Science.

Higher Secondary - 12th

Sri Narayani Vidyalaya

Jun 2012 - May 2013

- Relevant coursework in computer science.

Languages

- English (Fluent)
- Tamil (Fluent)
- Telugu (Proficient)
- kannada (Proficient)
- Hindi (Basic)

Certification/Short Courses

- Certified Backend Developer (CBD)
- Pursing on - Certified Data Scientist (CDA)

Additional Information

- Served as Team Lead in the company-wide initiative for process improvement.
- Local Nonprofit Organization.
- Received "Excellence" in Financial Management Award, Borcelle Corporation.

Tools/Software

- IDEs: VS Code, PyCharm
- Version Control: Git, GitHub, GitLab
- Project Management: Jira, Trello, Notion
- Containerization: Docker
- Visualization: Custom dashboards, Streamlit (basic)
- Monitoring: CloudWatch, basic ELK
- Collaboration: Slack, Zoom, MS Teams